📘 AI/ML Internship – Day 16 Notes & Tasks

🟢 Module 3 : Data Handling & Preparation

📌 Part 1: What is Data Preparation?

🔹 Definition

Data preparation means:
👉 Converting raw data into a clean and usable format

🔹 Steps in Data Preparation

1. Collect data
2. Clean data
3. Transform data
4. Prepare for analysis

📌 Part 2: Working with Structured Data

🔹 Example Dataset

```
students = [
    {"name": "A", "marks": 80},
    {"name": "B", "marks": None},
    {"name": "C", "marks": 40},
    {"name": "D", "marks": -10}
]
```

📌 Part 3: Data Cleaning (Advanced)

🔹 Remove Invalid Data

```
cleaned = []

for s in students:
    if s["marks"] is not None and s["marks"] >= 0:
        cleaned.append(s)
```

```
print(cleaned)
```

🔹 **Using List Comprehension**

```
cleaned = [s for s in students if s["marks"] is not None and s["marks"] >= 0]
```

📌 **Part 4: Handling Missing Values**

🔹 **Replace Missing Values**

```
for s in students:
    if s["marks"] is None:
        s["marks"] = 0
```

🔹 **Remove Missing Values**

```
students = [s for s in students if s["marks"] is not None]
```

📌 **Part 5: Data Transformation (Advanced)**

🔹 **Add New Field**

```
for s in students:
    s["status"] = "Pass" if s["marks"] >= 50 else "Fail"
```

🔹 **Normalize Data**

```
marks = [s["marks"] for s in students]

max_marks = max(marks)

normalized = [m / max_marks for m in marks]
print(normalized)
```

📌 **Part 6: Sorting Data**

```
students.sort(key=lambda x: x["marks"], reverse=True)
print(students)
```

📌 Part 7: Grouping Data

```
result = {"pass": [], "fail": []}

for s in students:
    if s["marks"] >= 50:
        result["pass"].append(s)
    else:
        result["fail"].append(s)

print(result)
```

📌 Part 8: Real-World Thinking

👉 This is exactly what happens before ML:

- Raw data → messy
- Clean → structured
- Transform → usable

🎯 Tasks for Day 16

🧠 Task 1: Theory

1. What is data preparation?
2. What is data cleaning?
3. What is missing data?
4. What is normalization?
5. Why data preparation is important?

💻 Task 2: Cleaning Practice

1. Remove None values
2. Remove negative values

3. Keep only valid data

💻 **Task 3: Transformation**

1. Add pass/fail field
2. Convert marks to percentage
3. Create new calculated field

💻 **Task 4: Sorting**

1. Sort students by marks
2. Print top 3 students

💻 **Task 5: Grouping**

1. Group pass and fail
2. Count each group

💻 **Task 6: Data Analysis**

1. Find average marks
2. Find highest marks
3. Find lowest marks